Selection and Mutation in a Fluctuating Environment:

A Hamilton-Jacobi Approach

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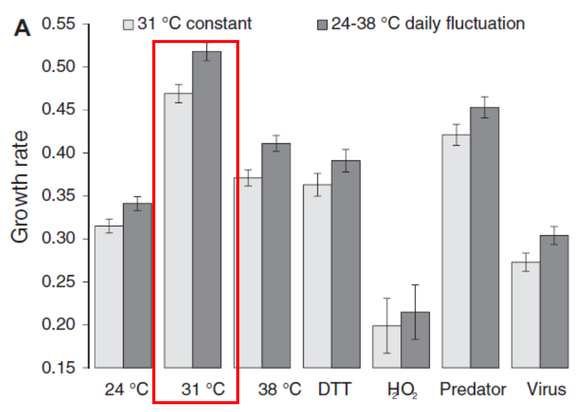
Motivation

Experiment: Several populations of of bacterial pathogen *Serratia marcescens* were kept in environments with constant or fluctuating temperature during several weeks and then their growth rates were measured in different environments.

Outcome: The population evolved in a periodic environment outperforms the one evolved at the constant environment, at the same constant environment. [Ketola et al. 2013]



Question: In which situations a population evolved in a periodic environment outperforms a population in a constant environment ?

The Model

Model:

$$\begin{cases} \partial_t n - \overbrace{\varepsilon^2 \Delta n}^{\text{Mutation}} = n\big[\overbrace{a(e(t),x)}^{\text{Growth}} - \overbrace{\rho(t)}^{\text{Competition}}\big], \\ \rho(t) = \int_{\mathbb{R}^d} n(t,x)dx, \\ n(t=0,x) = n_0(x). \end{cases}$$

Notations:

- $\mathbf{x}$: phenotypic trait
- $\mathbf{n}$: phenotypic density
- ε^2: effect of mutations
- $\rho(\mathbf{t})$: total size
- $\mathbf{a(e(t),x)}$: growth rate
- $\mathbf{e}$: environmental state

Main Assumptions:

- $e(t)$ is $T-$periodic
- a negative for x large
- $\max \bar{a}(x) > 0$ for $\bar{a}(x) = \frac{1}{T}\int_0^T a(t,x)dt$
- $n_0(x) \le e^{C_1 - C_2|x|}$

Long time behavior

Proposition [FI.-Mirrahimi, 2018]:
For small values of ε we have as $t \to \infty$,

$$|\rho(t) - \rho_\varepsilon(t)| \to 0, \|n(t,\cdot) - n_\varepsilon(t,\cdot)\|_{L^\infty} \to 0,$$

with $(n_\varepsilon, \rho_\varepsilon)$ the unique periodic solution of

$$\begin{cases} \partial_t n_\varepsilon - \varepsilon^2 \Delta n_\varepsilon = n_\varepsilon[a(e(t),x) - \rho_\varepsilon(t)], \\ \rho_\varepsilon(t) = \int_{\mathbb{R}^d} n_\varepsilon(t,x)dx, \quad n_\varepsilon(0,\cdot) = n_\varepsilon(T,\cdot). \end{cases}$$

Regime of small mutational effects ($\varepsilon << 1$):

Goal: Study $(n_\varepsilon, \rho_\varepsilon)$ asymptotically as $\varepsilon \to 0$
Tool: Hopf-Cole transformation:

$$n_\varepsilon = \frac{1}{(2\pi\varepsilon)^{d/2}} \exp\left(\frac{u_\varepsilon}{\varepsilon}\right).$$

Expected asymptotic expansion:

$$u_\varepsilon(t,x) = u + \varepsilon v + O(\varepsilon^2).$$

Asymptotic study of the phenotypic density

Theorem [FI.-Mirrahimi, 2018]:
Let x_m be the only point such that $\max \bar{a}(x) = \bar{a}(x_m)$. Then

(i) As $\varepsilon \to 0$: $n_\varepsilon(t,x) - \tilde{\varrho}(t)\delta(x - x_m) \rightharpoonup 0$, where $\tilde{\varrho}(t)$ is the unique periodic solution of

$$\frac{d\tilde{\varrho}}{dt} = \tilde{\varrho}[a(e(t),x_m) - \tilde{\varrho}].$$

(ii) As $\varepsilon \to 0$, $u_\varepsilon(t,x) = \varepsilon \ln n_\varepsilon(t,x)$ converges locally uniformly to $u(x)$, the unique viscosity solution of:

$$\begin{cases} -|\nabla u|^2 = \bar{a}(x) - \bar{a}(x_m), \quad x \in \mathbb{R}^d, \\ \max_{x\in\mathbb{R}^d} u(x) = u(x_m) = 0. \end{cases}$$

Heuristics
Let $u_\varepsilon = \varepsilon \ln n_\varepsilon$

$$\frac{1}{\varepsilon}\partial_t u_\varepsilon - \varepsilon \Delta u_\varepsilon = |\nabla u_\varepsilon|^2 + a - \rho_\varepsilon.$$

Asymptotic expansion for $u_\varepsilon, \rho_\varepsilon$ with $T-$periodic coefficients:

$$\begin{aligned} u_\varepsilon &= u(t,x) + \varepsilon v(t,x) + \varepsilon^2 w(t,x) + o(\varepsilon^2), \\ \rho_\varepsilon &= \rho(t) + \varepsilon \kappa(t) + o(\varepsilon), \end{aligned}$$

Substituting in the equation for u

$$\partial_t u = 0 \Rightarrow u(t,x) = u(x)$$

and

$$\partial_t v = |\nabla u|^2 + a(e(t),x) - \rho(t).$$

Integrating in $t \in [0,T]$

$$-|\nabla u|^2 = \bar{a}(x) - \bar{\rho}.$$

Idea of the proof
1. Regularity estimates based on comparison principle and Bernstein type method.

- Difficulty: time periodic condition.

2. Convergence to the limit equation using *a perturbed test function argument*.

3. Uniqueness of viscosity solution of the Hamilton-Jacobi equation thanks to:

- constraint of negativity on u
- $\max \bar{a}$ attained at a unique point

Biological examples

Notations

Phenotypic mean: $\mu(t) = \frac{1}{\rho_\varepsilon(t)}\int_{\mathbb{R}} x n_\varepsilon(t,x)dx$, Variance: $\sigma^2(t) = \frac{1}{\rho_\varepsilon(t)}\int_{\mathbb{R}}(x-\mu(t))^2 n_\varepsilon(t,x)dx$,

Mean Fitness: $F_{per}(\tau) = \int_{\mathbb{R}} a(\tau,x)\frac{1}{T}\int_0^T \frac{n_\varepsilon(t,x)}{\rho_\varepsilon(t)}dx$, $F_{cst} = \int_{\mathbb{R}} a(\tau,x)\frac{n_c(x)}{\rho_c}dx$

Oscillations on the optimal trait: $a(e(t),x) = r - g(x - c\sin bt)^2$

$$\underbrace{\mu(t) \approx \frac{2\varepsilon c\sqrt{g}}{b}\sin\left(bt - \frac{\pi}{2}\right)}_{\text{oscillates with the same period but with a delay and a smaller amplitude with respect to the optimal trait}}, \quad \underbrace{F_{per}(\frac{\pi}{b}) \approx r - \varepsilon\sqrt{g} = F_{cst}(\frac{\pi}{b}).}_{\text{same mean fitness for populations evolved either in a constant or fluctuating environment}}$$

Oscillations on the pressure of selection: $a(e(t),x) = r - g(e(t))x^2$, $g(e(t)) : 1-$ periodic.

$$\sigma^2_{per} \approx \frac{\varepsilon}{\sqrt{\bar{g}}} < \sigma^2_{cst} = \frac{\varepsilon}{\sqrt{g(1/2)}} \quad \text{and} \quad F_{per}(\tau) \approx r - \varepsilon\frac{g(1/2)}{\sqrt{\bar{g}}} > F_{cst} = r - \varepsilon\sqrt{g(1/2)}, \quad \text{if } \bar{g} > g(1/2)$$

Conclusion: in presence of the mutations and while the fluctuations act on the pressure of the selection, a fluctuating environment can select for a population with smaller variance and lead to more performant populations.

Extension

Changing environment with linear trend but in an oscillatory manner:
Growth rate: $\mathbf{a(e(t), x - ct)}$ with $\mathbf{c} > 0$.
Results:

- Approximation of the average size and the critical speed above which the population goes extinct.
- The phenotypic mean evolves with the same speed but with a constant lag with respect to the optimal trait, while the total size of the population oscillates periodically.

References

[1] Ketola, T. et al.(2013) Fluctuating temperature leads to evolution of thermal generalism and preadaptation to novel environments. *Evolution*, 67(10):2936-2944.

[2] F-I, S. & Mirrahimi, S. (2018). Long time evolutionary dynamics of phenotypically structured populations in time periodic environments, *SIAM Journal of Mathematical analysis*, 50(5):5537–5568.